

```
##
## Attaching package: 'dplyr'

## The following objects are masked from 'package:stats':
##
##   filter, lag

## The following objects are masked from 'package:base':
##
##   intersect, setdiff, setequal, union

##
## Attaching package: 'kableExtra'

## The following object is masked from 'package:dplyr':
##
##   group_rows

## Read Housing CSV file for houses
df_housing = read.csv("week-6-housing.csv")

## Selecting Sale.Date and Sale.Price columns from dataframe
year_country_gdp <- select(df_housing, Sale.Date, Sale.Price, sq_ft_lot)
head(year_country_gdp, 10)

##   Sale.Date Sale.Price sq_ft_lot
## 1   1/3/06     698000      6635
## 2   1/3/06     649990      5570
## 3   1/3/06     572500      8444
## 4   1/3/06     420000      9600
## 5   1/3/06     369900      7526
## 6   1/3/06     184667      7280
## 7   1/4/06    1050000     97574
## 8   1/4/06     875000     30649
## 9   1/4/06     660000     42688
## 10  1/4/06     650000     94889

demographicdata <- select(df_housing, ctyname, postalctyn)
head(demographicdata, 10)

##   ctyname postalctyn
## 1 REDMOND REDMOND
## 2 REDMOND REDMOND
## 3 REDMOND REDMOND
## 4 REDMOND REDMOND
## 5 REDMOND REDMOND
## 6 REDMOND REDMOND
## 7 REDMOND REDMOND
## 8 REDMOND REDMOND
## 9 REDMOND REDMOND
## 10 REDMOND REDMOND
```

```
## Filter
```

```
year_country_gdp_98052 <- filter(df_housing,zip5 == 98052)
```

```
year_country_gdp_98052_filtered <- select(year_country_gdp_98052,ctyname,postalctyn,year_built,sq_ft_lot,Sale.Price)
```

```
head(year_country_gdp_98052_filtered,20)
```

##	ctyname	postalctyn	year_built	sq_ft_lot	Sale.Price
## 1	REDMOND	REDMOND	2003	6635	698000
## 2	REDMOND	REDMOND	2006	5570	649990
## 3		REDMOND	1987	8444	572500
## 4	REDMOND	REDMOND	1968	9600	420000
## 5	REDMOND	REDMOND	1980	7526	369900
## 6	REDMOND	REDMOND	1976	94889	650000
## 7	REDMOND	REDMOND	1988	7949	599950
## 8	REDMOND	REDMOND	2005	2647	526787
## 9	REDMOND	REDMOND	1975	12070	470000
## 10	REDMOND	REDMOND	2005	3099	507950
## 11	REDMOND	REDMOND	2005	4737	589950
## 12	REDMOND	REDMOND	1972	9649	501000
## 13	REDMOND	REDMOND	1986	3279	372500
## 14	REDMOND	REDMOND	1998	17291	1392000
## 15	REDMOND	REDMOND	2006	5760	717390
## 16	REDMOND	REDMOND	1986	10827	552000
## 17	REDMOND	REDMOND	2005	3962	470000
## 18	REDMOND	REDMOND	1979	2752	335105
## 19	REDMOND	REDMOND	2006	3792	572950
## 20	REDMOND	REDMOND	2005	6204	949950

```
## cbind() and rbind() for binding dataframes
```

```
r_binding_df_20 = cbind(year_country_gdp,demographicdata)
```

```
head(r_binding_df_20,20)
```

##	Sale.Date	Sale.Price	sq_ft_lot	ctyname	postalctyn
## 1	1/3/06	698000	6635	REDMOND	REDMOND
## 2	1/3/06	649990	5570	REDMOND	REDMOND
## 3	1/3/06	572500	8444		REDMOND
## 4	1/3/06	420000	9600	REDMOND	REDMOND
## 5	1/3/06	369900	7526	REDMOND	REDMOND
## 6	1/3/06	184667	7280		REDMOND
## 7	1/4/06	1050000	97574		REDMOND
## 8	1/4/06	875000	30649		REDMOND
## 9	1/4/06	660000	42688		REDMOND
## 10	1/4/06	650000	94889	REDMOND	REDMOND
## 11	1/4/06	599950	7949	REDMOND	REDMOND
## 12	1/4/06	526787	2647	REDMOND	REDMOND
## 13	1/4/06	470000	12070	REDMOND	REDMOND
## 14	1/4/06	165000	278891		REDMOND
## 15	1/5/06	803000	95013		REDMOND
## 16	1/5/06	507950	3099	REDMOND	REDMOND
## 17	1/6/06	765000	7611		REDMOND
## 18	1/6/06	589950	4737	REDMOND	REDMOND
## 19	1/9/06	501000	9649	REDMOND	REDMOND
## 20	1/9/06	372500	47480		REDMOND

```
r_binding_df_30 = cbind(year_country_gdp,demographicdata)
head(r_binding_df_30,30)
```

```
##      Sale.Date Sale.Price sq_ft_lot ctyname postalctyn
## 1      1/3/06      698000      6635 REDMOND      REDMOND
## 2      1/3/06      649990      5570 REDMOND      REDMOND
## 3      1/3/06      572500      8444          REDMOND
## 4      1/3/06      420000      9600 REDMOND      REDMOND
## 5      1/3/06      369900      7526 REDMOND      REDMOND
## 6      1/3/06      184667      7280          REDMOND
## 7      1/4/06     1050000      97574          REDMOND
## 8      1/4/06      875000      30649          REDMOND
## 9      1/4/06      660000      42688          REDMOND
## 10     1/4/06      650000      94889 REDMOND      REDMOND
## 11     1/4/06      599950      7949 REDMOND      REDMOND
## 12     1/4/06      526787      2647 REDMOND      REDMOND
## 13     1/4/06      470000      12070 REDMOND      REDMOND
## 14     1/4/06      165000     278891          REDMOND
## 15     1/5/06      803000      95013          REDMOND
## 16     1/5/06      507950      3099 REDMOND      REDMOND
## 17     1/6/06      765000      7611          REDMOND
## 18     1/6/06      589950      4737 REDMOND      REDMOND
## 19     1/9/06      501000      9649 REDMOND      REDMOND
## 20     1/9/06      372500      47480          REDMOND
## 21     1/10/06     513262      4958          REDMOND
## 22     1/10/06     482000      4080          REDMOND
## 23     1/11/06      765000      35348          REDMOND
## 24     1/11/06      372500      3279 REDMOND      REDMOND
## 25     1/11/06      265000     112650          REDMOND
## 26     1/12/06     1392000      17291 REDMOND      REDMOND
## 27     1/12/06      717390      5760 REDMOND      REDMOND
## 28     1/12/06      552000      10827 REDMOND      REDMOND
## 29     1/12/06      470000      3962 REDMOND      REDMOND
## 30     1/13/06      523935      4764          REDMOND
```

```
r_binding_df = rbind(r_binding_df_20,r_binding_df_30)
head(r_binding_df,40)
```

```
##      Sale.Date Sale.Price sq_ft_lot ctyname postalctyn
## 1      1/3/06      698000      6635 REDMOND      REDMOND
## 2      1/3/06      649990      5570 REDMOND      REDMOND
## 3      1/3/06      572500      8444          REDMOND
## 4      1/3/06      420000      9600 REDMOND      REDMOND
## 5      1/3/06      369900      7526 REDMOND      REDMOND
## 6      1/3/06      184667      7280          REDMOND
## 7      1/4/06     1050000      97574          REDMOND
## 8      1/4/06      875000      30649          REDMOND
## 9      1/4/06      660000      42688          REDMOND
## 10     1/4/06      650000      94889 REDMOND      REDMOND
## 11     1/4/06      599950      7949 REDMOND      REDMOND
## 12     1/4/06      526787      2647 REDMOND      REDMOND
## 13     1/4/06      470000      12070 REDMOND      REDMOND
## 14     1/4/06      165000     278891          REDMOND
## 15     1/5/06      803000      95013          REDMOND
```

```
## 16 1/5/06 507950 3099 REDMOND REDMOND
## 17 1/6/06 765000 7611 REDMOND
## 18 1/6/06 589950 4737 REDMOND REDMOND
## 19 1/9/06 501000 9649 REDMOND REDMOND
## 20 1/9/06 372500 47480 REDMOND
## 21 1/10/06 513262 4958 REDMOND
## 22 1/10/06 482000 4080 REDMOND
## 23 1/11/06 765000 35348 REDMOND
## 24 1/11/06 372500 3279 REDMOND REDMOND
## 25 1/11/06 265000 112650 REDMOND
## 26 1/12/06 1392000 17291 REDMOND REDMOND
## 27 1/12/06 717390 5760 REDMOND REDMOND
## 28 1/12/06 552000 10827 REDMOND REDMOND
## 29 1/12/06 470000 3962 REDMOND REDMOND
## 30 1/13/06 523935 4764 REDMOND
## 31 1/13/06 399900 4781 REDMOND
## 32 1/13/06 335105 2752 REDMOND REDMOND
## 33 1/16/06 572950 3792 REDMOND REDMOND
## 34 1/17/06 949950 6204 REDMOND REDMOND
## 35 1/17/06 905000 46270 REDMOND
## 36 1/17/06 750073 5697 REDMOND
## 37 1/17/06 526718 6386 REDMOND
## 38 1/18/06 690749 7583 REDMOND
## 39 1/18/06 640000 35006 REDMOND
## 40 1/18/06 485000 7560 REDMOND
```

```
## GroupBy and Summarise sq_ft_lot by Zip
```

```
df_Group_by_zip<-group_by(df_housing,zip5)%>%summarise(TotalSqFt=sum(sq_ft_lot))
df_Group_by_zip
```

```
## # A tibble: 4 x 2
##   zip5 TotalSqFt
##   <int>     <int>
## 1 98052  89123232
## 2 98053 193522888
## 3 98059   39109
## 4 98074  3285301
```

```
df_mutate <- mutate(year_country_gdp_98052_filtered
,price_per_sqft = Sale.Price/sq_ft_lot)
head(df_mutate,20)
```

```
##   ctyname postalctyn year_built sq_ft_lot Sale.Price price_per_sqft
## 1 REDMOND REDMOND 2003 6635 698000 105.199699
## 2 REDMOND REDMOND 2006 5570 649990 116.694794
## 3 REDMOND REDMOND 1987 8444 572500 67.799621
## 4 REDMOND REDMOND 1968 9600 420000 43.750000
## 5 REDMOND REDMOND 1980 7526 369900 49.149615
## 6 REDMOND REDMOND 1976 94889 650000 6.850109
## 7 REDMOND REDMOND 1988 7949 599950 75.474903
## 8 REDMOND REDMOND 2005 2647 526787 199.012845
## 9 REDMOND REDMOND 1975 12070 470000 38.939519
## 10 REDMOND REDMOND 2005 3099 507950 163.907712
## 11 REDMOND REDMOND 2005 4737 589950 124.540849
## 12 REDMOND REDMOND 1972 9649 501000 51.922479
```

```
## 13 REDMOND REDMOND 1986 3279 372500 113.601708
## 14 REDMOND REDMOND 1998 17291 1392000 80.504309
## 15 REDMOND REDMOND 2006 5760 717390 124.546875
## 16 REDMOND REDMOND 1986 10827 552000 50.983652
## 17 REDMOND REDMOND 2005 3962 470000 118.626956
## 18 REDMOND REDMOND 1979 2752 335105 121.767805
## 19 REDMOND REDMOND 2006 3792 572950 151.094409
## 20 REDMOND REDMOND 2005 6204 949950 153.118956
```

```
## Arrange data frame by sq_ft_lot
```

```
ar_r_df_housing <- head(r_binding_df,10)
arrange(ar_r_df_housing,desc(sq_ft_lot))
```

```
## Sale.Date Sale.Price sq_ft_lot ctyname postalctyn
## 1 1/4/06 1050000 97574 REDMOND REDMOND
## 2 1/4/06 650000 94889 REDMOND REDMOND
## 3 1/4/06 660000 42688 REDMOND REDMOND
## 4 1/4/06 875000 30649 REDMOND REDMOND
## 5 1/3/06 420000 9600 REDMOND REDMOND
## 6 1/3/06 572500 8444 REDMOND REDMOND
## 7 1/3/06 369900 7526 REDMOND REDMOND
## 8 1/3/06 184667 7280 REDMOND REDMOND
## 9 1/3/06 698000 6635 REDMOND REDMOND
## 10 1/3/06 649990 5570 REDMOND REDMOND
```

```
## keep and discard functions for dataframe values
```

```
df_discard_values <- discard(ar_r_df_housing,is.character)
df_discard_values
```

```
## Sale.Price sq_ft_lot
## 1 698000 6635
## 2 649990 5570
## 3 572500 8444
## 4 420000 9600
## 5 369900 7526
## 6 184667 7280
## 7 1050000 97574
## 8 875000 30649
## 9 660000 42688
## 10 650000 94889
```

```
df_keep_values <- keep(ar_r_df_housing,is.character)
df_keep_values
```

```
## Sale.Date ctyname postalctyn
## 1 1/3/06 REDMOND REDMOND
## 2 1/3/06 REDMOND REDMOND
## 3 1/3/06 REDMOND REDMOND
## 4 1/3/06 REDMOND REDMOND
## 5 1/3/06 REDMOND REDMOND
## 6 1/3/06 REDMOND REDMOND
## 7 1/4/06 REDMOND REDMOND
## 8 1/4/06 REDMOND REDMOND
## 9 1/4/06 REDMOND REDMOND
## 10 1/4/06 REDMOND REDMOND
```

```

## Split a string and Combined it altogether
addr_df <- select(df_housing, addr_full)
addr_df_10 <- head(addr_df, 2)

df_addr_split <- strsplit(as.character(addr_df_10), split=" ")
df_addr_split

## [[1]]
## [1] "c(\"17021\" \"NE\" \"113TH\" \"CT\", \" \"11927\" \"178TH\"
## [7] \"PL\" \"NE\")"

df_addr_concat <- paste0(as.character(df_housing$addr_full), " ",
                          as.character(df_housing$zip5))
head(df_addr_concat, 10)

## [1] "17021 NE 113TH CT 98052" "11927 178TH PL NE 98052"
## [3] "13315 174TH AVE NE 98052" "3303 178TH AVE NE 98052"
## [5] "16126 NE 108TH CT 98052" "8101 229TH DR NE 98053"
## [7] "21634 NE 87TH PL 98053" "21404 NE 67TH ST 98053"
## [9] "7525 238TH AVE NE 98053" "17703 NE 26TH ST 98052"

```